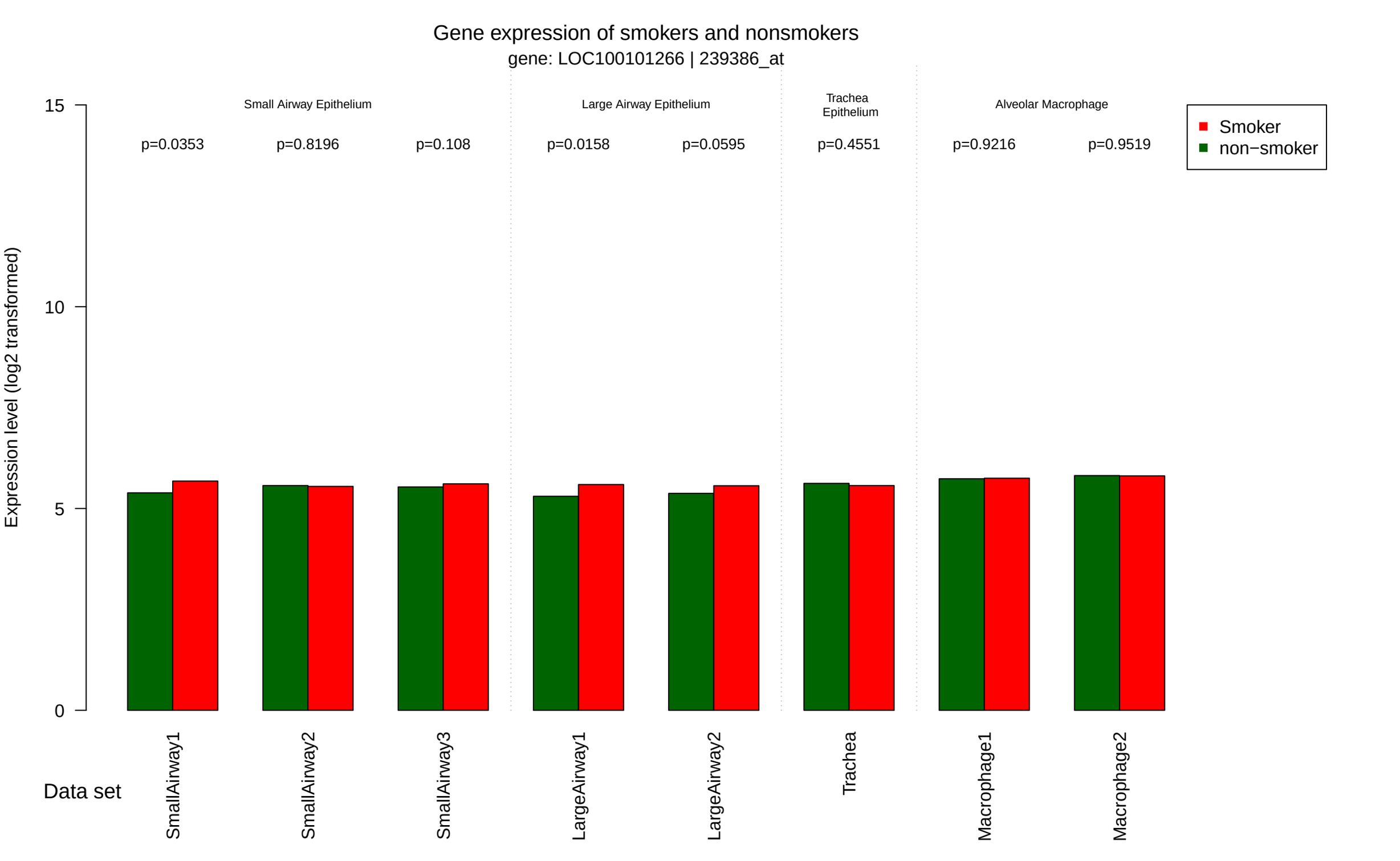
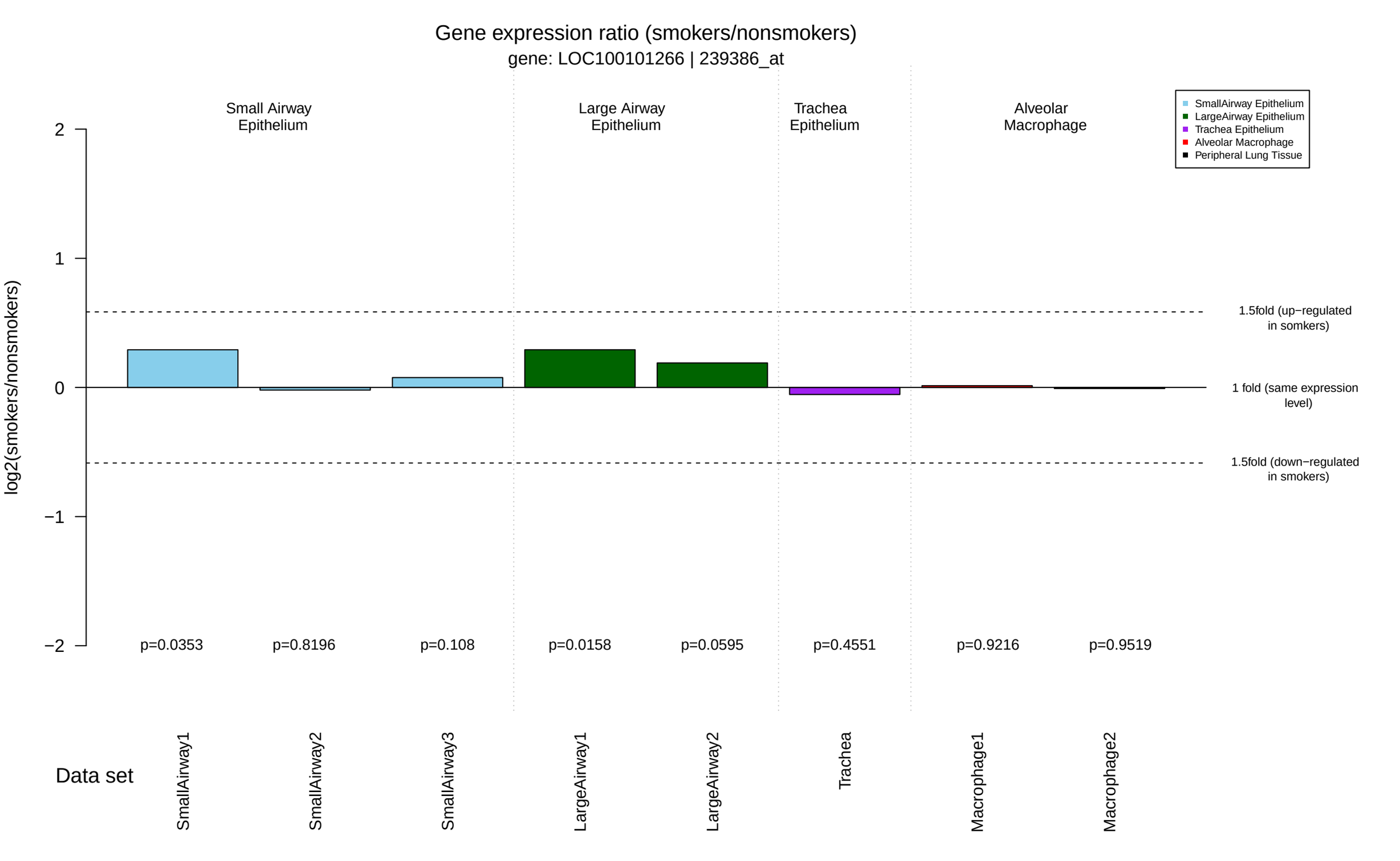
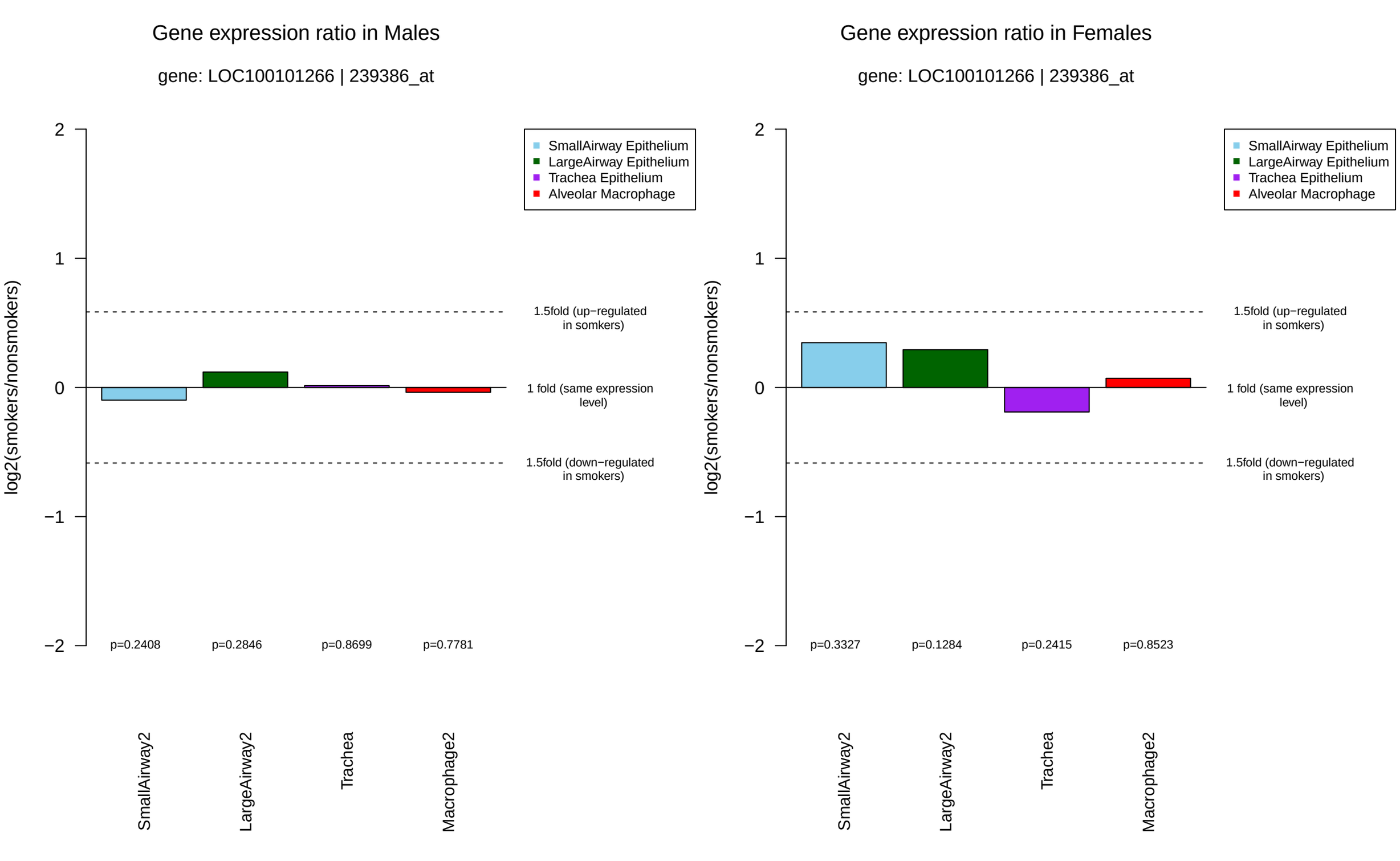
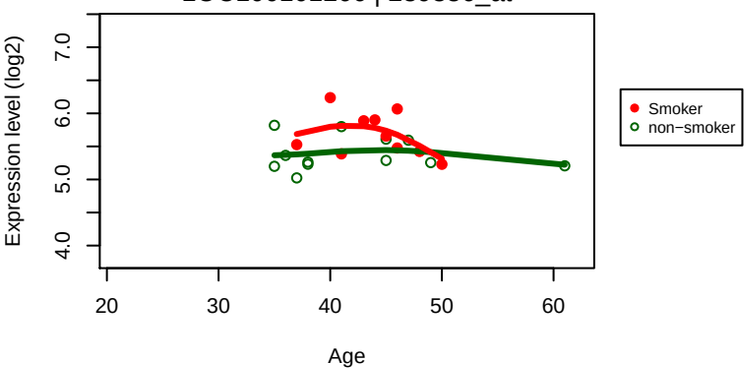
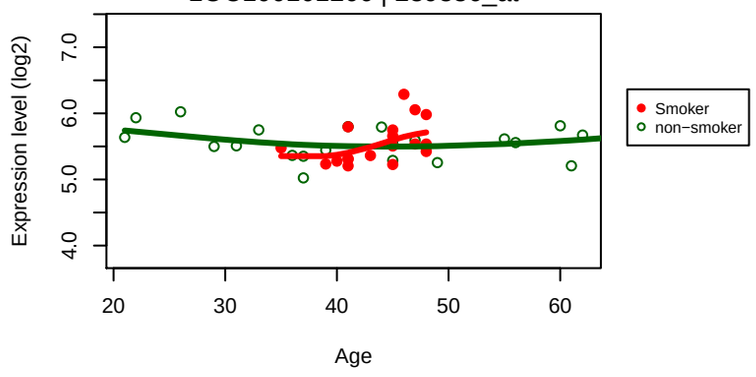
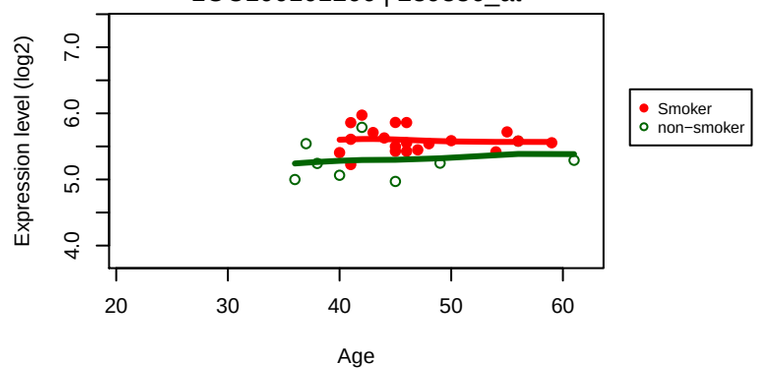

Gene expression of smokers and nonsmokers
gene: LOC100101266 | 239386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0353
p=0.8196
p=0.108
p=0.0158
p=0.0595
p=0.4551
p=0.9216
p=0.9519
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

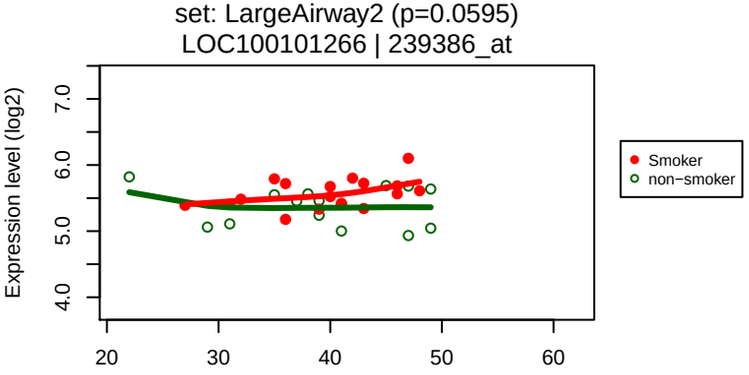

Gene expression ratio (smokers/nonsmokers)
gene: LOC100101266 | 239386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0353
p=0.8196
p=0.108
p=0.0158
p=0.0595
p=0.4551
p=0.9216
p=0.9519
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

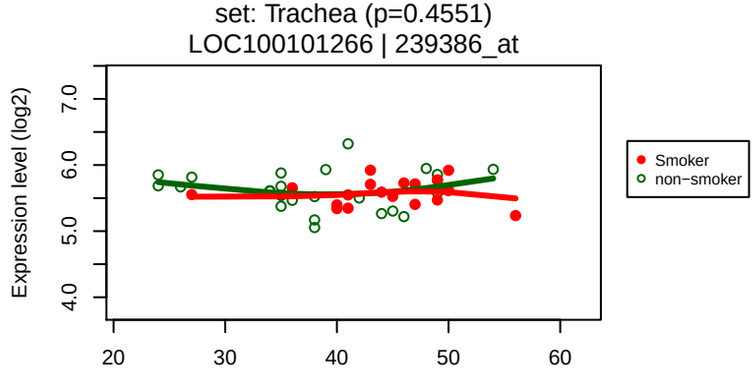

Gene expression ratio in Males
gene: LOC100101266 | 239386_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2408
p=0.2846
p=0.8699
p=0.7781
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100101266 | 239386_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3327
p=0.1284
p=0.2415
p=0.8523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

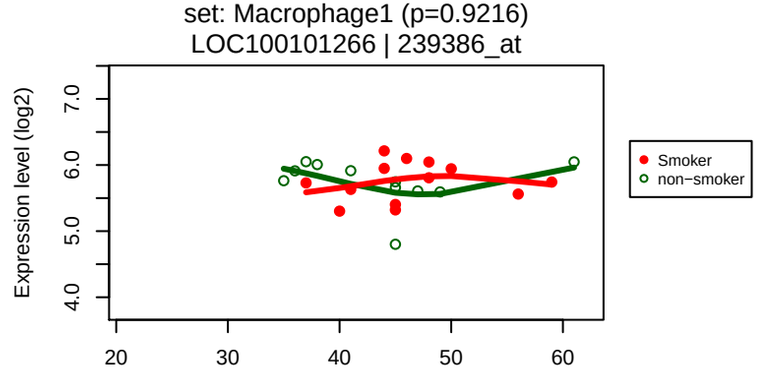

set: SmallAirway1 (p=0.0353)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

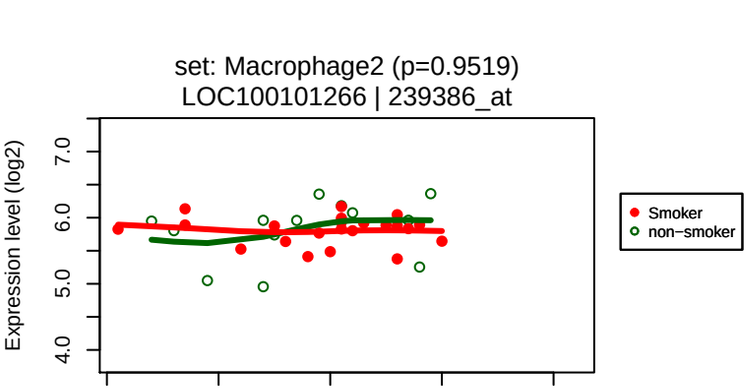

set: SmallAirway2 (p=0.8196)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

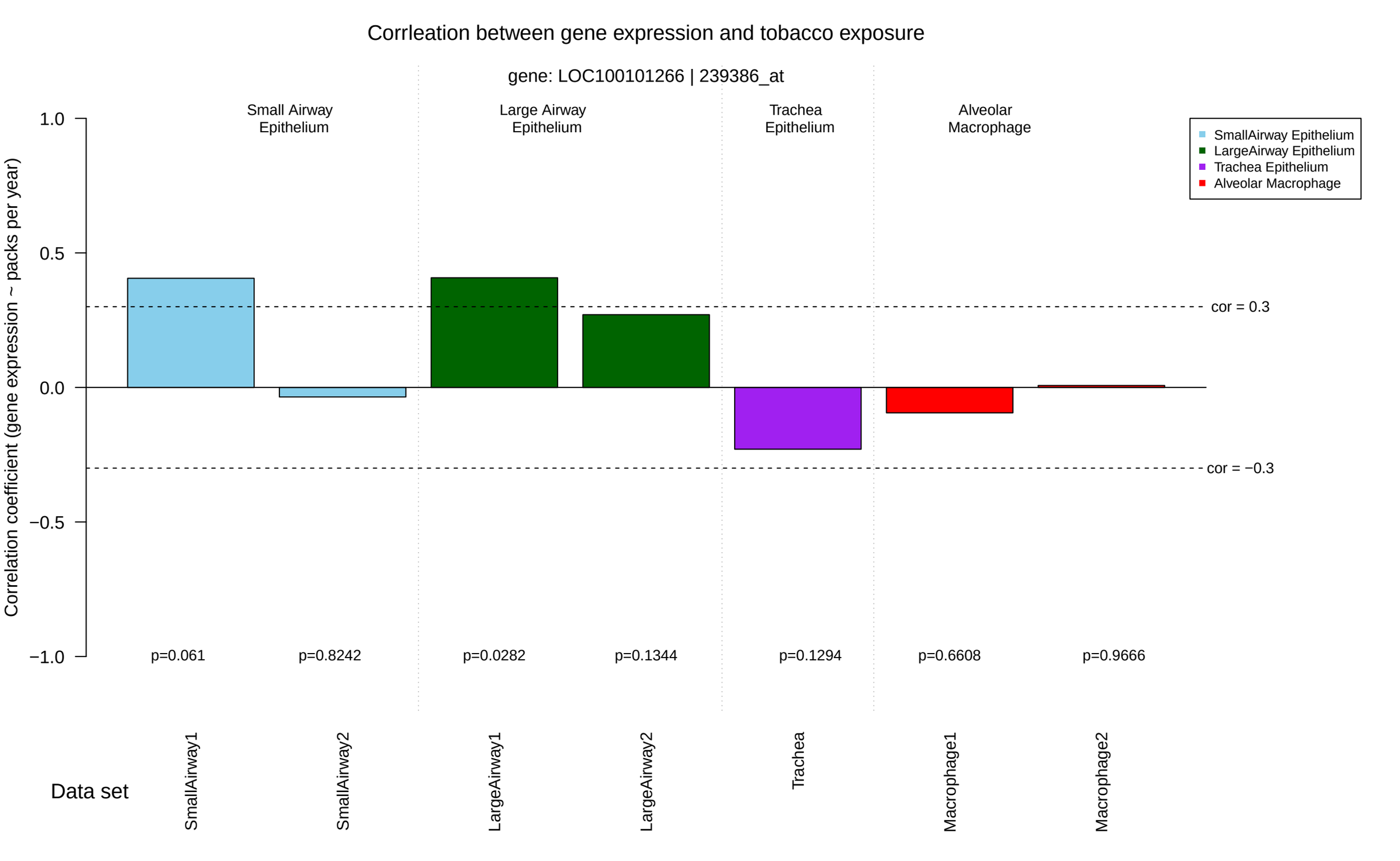

set: LargeAirway1 (p=0.0158)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0595)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4551)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9216)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9519)
LOC100101266 | 239386_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100101266 | 239386_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.061
p=0.8242
p=0.0282
p=0.1344
p=0.1294
p=0.6608
p=0.9666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2